1 The Jordan-Wigner mapping for spinless Fermions

1.1 Summary

The Jordan-Wigner transformation (JWT) can be used to map a fermionic system to a system of spins. This is essentially done by rewriting

$$\text{fermion} = \text{string} \times \text{operator},$$

where the "string" is some product of operators $\hat{Z}_i = 1 - 2\hat{n}_i$ and the "operator" is found to obey spin commutation relations. Within this section, we only consider spinless fermions, but the JWT can also be applied on spinful systems.

The importance of the JWT can be seen, when examining how to compute some general matrix element $\langle x| \hat{A} |x'\rangle$, where the operator $\hat{A}$ is given as some product of fermionic creation and annihilation operators. To determine the matrix element, one has to commute operators and keep track of the resulting minus signs. This is exactly what the Jordan-Wigner mapping does for us.

Refer to **??** to see what role the JWT plays in variational Monte Carlo.

1.2 Setup: the fermionic Hilbert space

Our fermionic Hilbert space is defined by (Nielsen):

i) operators $c_1, \ldots, c_L$ acting on some Hilbert space;

ii) these operators satisfying the canonical (anti-)commutation relations for fermions

$$\{c_i^{(+)}, c_j^{(+)}\} = 0 \qquad \& \qquad \{c_i, c_j^\dagger\} = \delta_{ij}\, \mathbb{1}. \tag{1}$$

Many things directly follow from this, such as a possible basis for the Hilbert space, namely the occupation number basis

$$|\boldsymbol{n}\rangle = |n_1, n_2, \ldots, n_L\rangle \equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right) |0\rangle\,, \tag{2}$$

where $\hat{n}_i |\boldsymbol{n}\rangle = c_i^\dagger c_i |\boldsymbol{n}\rangle = n_i |\boldsymbol{n}\rangle$ with $n_i = 0, 1$ and

$$[\hat{n}_i, \hat{n}_j] = 0. \tag{3}$$

1.3 The transformation

The JWT introduces new operators a_j by writing

$$c_j \equiv \left(\prod_{i=1}^{j-1} Z_i\right) a_j \qquad \& \qquad c_j^\dagger \equiv a_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right), \tag{4}$$

with $Z_j = (-1)^{n_j} = 1 - 2n_j$ and $n_j = c_j^\dagger c_j$. The ordering of the Z_i does not matter since $[Z_i, Z_j] = 0$ due to eq. (3). Inverting eq. (4) using $Z_i^2 = \mathbb{1}$ and $[Z_i, Z_j] = 0$ yields

$$a_j = \left(\prod_{i=1}^{j-1} Z_i\right) c_j \qquad \& \qquad a_j^\dagger = c_j^\dagger \left(\prod_{i=1}^{j-1} Z_i\right). \tag{5}$$

1.4 Algebra of the new operators

One can show that these new operators obey

$$[a_i^{(+)}, a_j^{(+)}] = 0 \qquad \forall\, i, j, \tag{6}$$

$$[a_i, a_j^\dagger] = 0 \qquad \forall\, i \neq j, \tag{7}$$

$$[a_i, a_i^\dagger] = 1 - 2n_i \qquad \text{with } n_i = c_i^\dagger c_i, \tag{8}$$

using that

$$\{Z_i, c_j^{(+)}\} = 0 \tag{9}$$

The corresponding proofs can be found in section 1.8.

1.5 The occupation-number basis in the new operators

The basis states of the fermionic Hilbert space can be written as

$$\begin{aligned}
|n_1, n_2, \ldots, n_L\rangle &\equiv \left(\prod_{i=1}^{L} (c_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1 \\
&= \left\{\prod_{i=1}^{L} \left[a_i^\dagger \left(\prod_{j=1}^{i-1} Z_j\right)\right]^{n_i}\right\} |0\rangle && (c_1^\dagger = a_1^\dagger) \\
&= (a_1^\dagger)^{n_1} (a_2^\dagger Z_1)^{n_2} (a_3^\dagger Z_2 Z_1)^{n_3} \cdots (a_L^\dagger Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) (Z_1)^{n_2} (Z_2 Z_1)^{n_3} \cdots (Z_{L-1} \cdots Z_1)^{n_L} |0\rangle \\
&= \left(\prod_{i=1}^{L} (a_i^\dagger)^{n_i}\right) |0\rangle && \text{w. } n_i = 0, 1,
\end{aligned} \tag{10}$$

using $[Z_i, Z_j] = 0\ \forall i, j$, $[Z_i, c_j] = 0$ for $i \neq j$ (hence $[Z_i, a_j^\dagger] = 0$ for $i \neq j$), and $Z_j |0\rangle = |0\rangle$. The ordering is not important anymore because $[a_i^\dagger, a_j^\dagger] = 0$.

1.6 Action on basis states

One cannot define the action of $a_j\,(a_j^\dagger)$ locally, since the JW string $Z_1\cdots Z_{j-1}$ is non-local, so one has to examine them on the full many-body basis. For $n_\ell = 0$,

$$\begin{aligned} a_\ell\,|n_1,\ldots,n_\ell=0,\ldots,n_L\rangle &= a_\ell\left(\prod_{i=1}^{L}(a_i^\dagger)^{n_i}\right)|0\rangle \\ &\overset{[a_\ell,a_j^\dagger]=0}{=} \left(\prod_{i=1}^{L}(a_i^\dagger)^{n_i}\right) a_\ell\,|0\rangle = 0, \end{aligned} \tag{11}$$

since $a_\ell\,|0\rangle = c_\ell\,|0\rangle = 0$. For $n_\ell = 1$,

$$\begin{aligned} a_\ell\,|n_1,\ldots,n_\ell=1,\ldots,n_L\rangle &= a_\ell\left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right) a_\ell^\dagger\,|0\rangle \\ &\overset{[a_\ell,a_j^\dagger]=0}{=} \left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right) a_\ell a_\ell^\dagger\,|0\rangle = \left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right) c_\ell c_\ell^\dagger\,|0\rangle \\ &= \left(\prod_{\substack{i=1\\ i\neq\ell}}^{L}(a_i^\dagger)^{n_i}\right)|0\rangle = |n_1,\ldots,n_\ell=0,\ldots,n_L\rangle\,. \end{aligned} \tag{12}$$

Using $[a_\ell^\dagger, a_j^\dagger] = 0$ and $(a_\ell^\dagger)^2 = 0$, one quickly finds

$$a_\ell^\dagger\,|n_1,\ldots,n_\ell=0,\ldots,n_L\rangle = |n_1,\ldots,n_\ell=1,\ldots,n_L\rangle\,, \tag{13}$$

$$a_\ell^\dagger\,|n_1,\ldots,n_\ell=1,\ldots,n_L\rangle = 0. \tag{14}$$

Thus $a_\ell\,(a_\ell^\dagger)$ destroys (creates) a particle in the many-body state *without* giving rise to an extra sign structure (as $c_\ell\,(c_\ell^\dagger)$ would do).

1.7 Identification with spins

We can thus identify $n_i = 0, 1$ with $n_i = \downarrow, \uparrow$ in the many-body state, and the new operators can be associated with spins by identifying

$$a_j = S_j^-, \qquad a_j^\dagger = S_j^+ \qquad \& \qquad n_j - \tfrac{1}{2} = S_j^z = \tfrac{1}{2}\,\sigma_j^z, \tag{15}$$

since $[S_j^-, S_j^+] = -2S_j^z$ for spins (see the quick refresher on spin, **??**). For instance

$$a_3^\dagger\,|0\rangle = S_3^+\,|\downarrow\downarrow\downarrow\rangle = |\uparrow\downarrow\downarrow\rangle\,.$$

Matrix elements do not change under the JWT, since it is only a rewriting of operators and a renaming of basis states.

1.8 Proofs about the JWT

Claim: $\{Z_j, c_j^{(+)}\} = 0$**.**

Proof. On the single site j, with $Z_j |0\rangle_j = |0\rangle_j$, $Z_j |1\rangle_j = -|1\rangle_j$, $c_j |0\rangle_j = 0$ and $c_j |1\rangle_j = |0\rangle_j$,

$$Z_j c_j |0\rangle_j = Z_j \cdot 0 = 0 = c_j |0\rangle_j = c_j Z_j |0\rangle_j , \tag{16}$$

$$Z_j c_j |1\rangle_j = Z_j |0\rangle_j = |0\rangle_j = c_j |1\rangle_j , \tag{17}$$

$$c_j Z_j |1\rangle_j = -c_j |1\rangle_j = -|0\rangle_j . \tag{18}$$

Hence, on both basis states $\{Z_j, c_j\} = Z_j c_j + c_j Z_j = c_j - c_j = 0$, and since $Z_j^\dagger = Z_j$ it follows that $\{Z_j, c_j^\dagger\} = 0$ as well. (see Coleman, Chapter 4.1) □

Claim: $[a_i^{(+)}, a_j^{(+)}] = 0 \;\; \forall\, i, j$**.**

Proof. Assume w.l.o.g. that $i < j$. Using eq. (5),

$$a_j a_i = (Z_1 \cdots Z_{j-1} c_j)(Z_1 \cdots Z_{i-1} c_i)$$
$$\overset{\substack{[Z_j, c_\ell]=0,\, j\neq\ell \\ \{c_i, c_j\}=0}}{=} Z_1^2 \cdots Z_{i-1}^2 \, Z_i \cdots Z_{j-1} \, c_j c_i = -Z_i \cdots Z_{j-1} \, c_i c_j \qquad (= 0 \text{ if } i = j), \tag{19}$$

and likewise

$$a_i a_j = (Z_1 \cdots Z_{i-1} c_i)(Z_1 \cdots Z_{j-1} c_j)$$
$$\overset{[Z_i, c_\ell]=0,\, i\neq\ell}{=} Z_{i+1} \cdots Z_{j-1} \, c_i Z_i c_j \overset{\{Z_i, c_i\}=0}{=} -Z_i \cdots Z_{j-1} \, c_i c_j . \tag{20}$$

Comparing the two, $a_i a_j = a_j a_i$, i.e. $[a_i, a_j] = 0$. Analogously for $[a_i^\dagger, a_j^\dagger]$ and for $i > j$; the case $i = j$ is straightforward since $c_i^2 = 0$. □

Claim: $[a_i, a_j^\dagger] = 0 \;\; \forall\, i \neq j$**.**

Proof. We can show it w.l.o.g. for $i < j$. Using eq. (5) and eq. (9),

$$a_i^\dagger a_j = \big(c_i^\dagger Z_{i-1} \cdots Z_1\big)\big(Z_1 \cdots Z_{j-1} c_j\big) \overset{Z_i^2=1,\, i<j}{=} c_i^\dagger \, Z_i \cdots Z_{j-1} \, c_j$$
$$= Z_{i+1} \cdots Z_{j-1} \, c_i^\dagger Z_i c_j \overset{\{Z_i, c_i^\dagger\}=0}{=} -Z_i \cdots Z_{j-1} \, c_i^\dagger c_j = +Z_i \cdots Z_{j-1} \, c_j c_i^\dagger , \tag{21}$$

and

$$a_j a_i^\dagger = \big(Z_1 \cdots Z_{j-1} c_j\big)\big(c_i^\dagger Z_{i-1} \cdots Z_1\big)$$
$$= Z_i \cdots Z_{j-1} \, c_j c_i^\dagger \, Z_{i-1}^2 \cdots Z_1^2 \qquad ([Z_j, c_\ell^{(+)}] = 0,\; j \neq \ell)$$
$$= Z_i \cdots Z_{j-1} \, c_j c_i^\dagger . \tag{22}$$

Both reduce to the same expression, so $a_j a_i^\dagger - a_i^\dagger a_j = 0$ for $i < j$, i.e. $[a_j, a_i^\dagger] = 0$; the case $i > j$ follows straightforwardly. □

Claim: $[a_i, a_i^\dagger] = 1 - 2n_i$ **with** $n_i = c_i^\dagger c_i$**.**

Proof. Using eq. (5) and $Z_k^2 = \mathbb{1}$,

$$a_i^\dagger a_i = c_i^\dagger \, Z_{i-1} \cdots Z_1 Z_1 \cdots Z_{i-1} \, c_i = c_i^\dagger c_i = n_i, \tag{23}$$

$$a_i a_i^\dagger = Z_1 \cdots Z_{i-1} \, c_i c_i^\dagger \, Z_{i-1} \cdots Z_1 = c_i c_i^\dagger = 1 - c_i^\dagger c_i, \tag{24}$$

so that

$$[\hat{a}_i, \hat{a}_i^\dagger] = a_i a_i^\dagger - a_i^\dagger a_i = (1 - c_i^\dagger c_i) - c_i^\dagger c_i = 1 - 2\hat{n}_i. \tag{25}$$

□

1.9 References

- *Fermions and Jordan-Wigner String* - ITensor C++ Documentation,
- *Introduction to Many-Body Physics*, Chapter 4.1 - Piers Coleman,
- *The Fermionic canonical commutation relations and the Jordan-Wigner transform* - M. Nielsen,